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What is an Integer Linear Program?

NP-Hardness of Integer Linear Programs

Tutorial: Solving Integer Linear Programs using Python/Pulp

✔ Lab: Tutorial on solving ILPs using Python/PuLP package 1h

✔ Quiz: Formulating/Solving ILPs Submitted

Vertex Cover as Integer Linear Program

Linear Programming Approximations of Vertex Cover

Solving Integer Linear Programs using Branch and Bound Algorithm

Problem Set # 2

Formulating/Solving ILPs

Review Learning Objectives

✔ Submit your assignment

Due Feb 18, 11:59 PM IST

✔ Receive grade

To Pass 80% or higher

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✔ Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

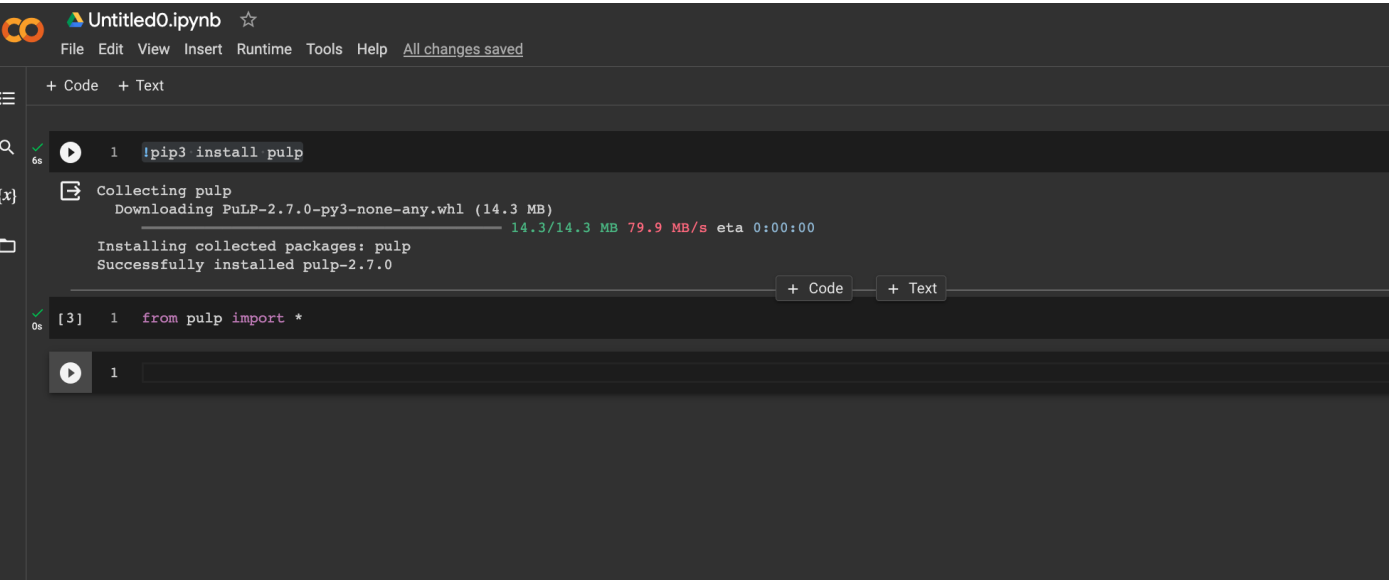
To pass 80% or higher

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Important Note

The problems below ask you to setup and solve ILPs using Pulp. It helps to have python3 installed in your computer. If you have a package manager such as pip/conda, you can directly use them to install pulp library. This will allow you to program in python either through your REPL, or through an IDE such as Atom, VSCode, IntelliJ PyCharm and so on.

Alternatively, if you are not setup to program on your local machine, we suggest using [google.colab](#) to have a python3 jupyter notebook. In the very first cell of the notebook type `!pip3 install pulp` to install the pulp library. In the next cell please type `from pulp import *` to check if it got installed.



You can type python programs to solve the ILPs and find out the answers for the problems below.

The questions below pertain to the ILP

$$\begin{aligned} \max \quad & 2x_1 - 3x_2 + x_3 \\ \text{s.t.} \quad & x_1 - x_2 + x_3 \leq 5 \\ & x_1 - x_2 + 4x_3 \leq 7 \\ & x_1 + 2x_2 - x_3 + x_4 \leq 14 \\ & x_3 - x_4 + x_5 \leq 7 \\ & x_1, x_2, x_3, x_4, x_5 \in [-15, 15] \\ & x_1, x_2, x_3, x_4, x_5 \in \mathbb{Z} \end{aligned}$$

1. Please use a python interpreter to setup and solve the following integer linear programming problem using pulp.

3 / 3 points

$$\begin{aligned} \max \quad & 2x_1 - 3x_2 + x_3 \\ \text{s.t.} \quad & x_1 - x_2 + x_3 \leq 5 \\ & x_1 - x_2 + 4x_3 \leq 7 \\ & x_1 + 2x_2 - x_3 + x_4 \leq 14 \\ & x_3 - x_4 + x_5 \leq 7 \\ & x_1, x_2, x_3, x_4, x_5 \in [-15, 15] \\ & x_1, x_2, x_3, x_4, x_5 \in \mathbb{Z} \end{aligned}$$

Write down the optimal value of the objective function correct to two decimal places.

40

✔ Correct
Correct

2. Solve the LP relaxation of the ILP from the previous problem using PuLP. What is the answer obtained?

2 / 2 points

40

✔ Correct

3. What can you conclude from solving the LP relaxation and original ILP for the previous problem.

3 / 3 points

- ☒ The integrality gap is 1.
- ✔ Correct

Indeed: we find that both the original ILP and LP relaxation are close together.
- ☐ The ILP solution is strictly larger than the LP relaxation.
- ☒ They have the same solution.
- ✔ Correct

That is what we got as well.
- ☒ The LP relaxation yields an integral solution for all the decision variables and we know from just that information that the ILP will also have the same optimal solution.
- ✔ Correct

Correct: if the LP relaxation yields an integral optimal solution, that is also the solution to the ILP.
- ☐ The integrality gap for this problem is 2.

The questions below pertains to the following ILP

$$\begin{aligned} \min \quad & 2x_1 - 3x_2 + x_3 \\ \text{s.t.} \quad & x_1 - x_2 \geq 0.5 \\ & x_1 - x_2 \leq 0.75 \\ & x_2 - x_3 \leq 1.25 \\ & x_2 - x_3 \geq 0.95 \\ & x_1, x_2, x_3 \in [-1, 1] \\ & x_1, x_2, x_3 \in \mathbb{Z} \end{aligned}$$

4. What is the status obtained when solving the ILP?

1 / 1 point

- ☐ Unbounded
- ☐ Optimal solution obtained.
- ☒ Infeasible
- ✔ Correct

Correct

5. What is the status of the LP relaxation?

1 / 1 point

- ☒ Optimal solution obtained
- ☐ Unbounded
- ☐ Infeasible
- ✔ Correct

Correct

